HHH-V-002133(Army-GL)
October 5, 1983

INTERIM FEDERAL SPECIFICATION

PEAS AND MUSHROOMS, THERMOSTABILIZED, TRAY PACK

This interim Federal specification was developed by U.S. Army Natick Research and Development Laboratories, Natick, MA 01760, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers peas and mushrooms thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards and Commercial Item Descrtiption from established distribution points in their agencies.)

FSC 8915

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Military Specifications:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence Items; General Specifications for

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Canned Foods for the Armed Forces

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

U.S. Department of Health and Human Services

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Department of Agriculture (USDA)

United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Food Quality Assurance Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

Regulations Governing Inspection and Certification of Processed Fruits and Vegetables, Processed Products Thereof, and Certain Other Processed Food Products

U.S. Standards for Grades of Frozen Peas
U.S. Standards for Grades of Canned Peas
U.S. Standards for Grades of Canned Mushrooms

(Application for copies should be addressed to the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service (AMS), U.S. Department of Agriculture, Washington, DC 20250.)

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3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.1 and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Peas. The peas shall be fresh, clean, sound, succulent seed of a canning variety (Alaskas or No. 3 extra standard sweets) or U.S. Grade A frozen peas conforming to the U.S. Standards for Grades of Frozen Peas of the latest season's pack.

3.2.2 Mushrooms, sliced, canned. The sliced mushrooms shall be prepared from clean, sound, edible mushrooms. The mushrooms shall conform to U.S. Grade A of the U.S. Standards for Grades of Canned Mushrooms of the latest season's pack.

3.2.3 Salt. Salt shall be white, refined, sodium chloride with or without anticaking agent. Iodized salt shall not be used.

3.2.4 Sugar. Sugar shall be white, refined, granulated cane or beet sugar, or a combination thereof. It shall meet the bacterial requirements of MIL-STD-900.

3.3 Preparation and processing.

3.3.1 Peas preparation. The fresh peas shall be put through a dry cleaner, washed, screened, and cleaned to remove pieces of sticks, pods, stones, empty skins, thistle buds, or other extraneous material present. The fresh peas shall be blanched for approximately 3 minutes in near boiling water (190° to 205°F). The peas shall be of such size that not more than 5 percent, by weight are less than 10/32 inch in diameter nor more than 5 percent are greater than 14/32 inch in diameter. Previously blanched frozen peas shall not be reblanched.

3.3.2 Mushroom preparation. The canned, sliced mushrooms shall be thoroughly drained prior to mixing with the peas.

3.3.3 Brine preparation. The brine shall consist of 15 pounds salt and 25 pounds sugar per 100 gallons of water. The brine shall be heated to at least 200°F prior to adding to the product.

3.3.4 Mixing peas and mushrooms. Peas and mushrooms shall be mixed in a weight ratio of four parts peas to one part mushrooms. Any method for mixing may be used that guarantees a uniform blend and prevents crushing of the peas.

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3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with peas, mushrooms, and brine such as to conform to the following finished product requirements:

a. The average net weight shall be 108 ounces minimum.
b. No individual can shall contain less than 106 ounces of product.
c. No individual can shall contain less than 71 ounces drained weight of peas and mushrooms.

The temperature of the brine at time of filling shall be at least 200°F. The cans shall be hermetically sealed with 10-15 inches of vacuum immediately after filling.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 7.2 has been achieved.

3.6 Finished product. The finished product shall meet the requirements for U.S. Grade A of the United States Standards for Grades of Canned Peas and U.S. Standards for Grades of Canned Mushrooms.

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed, and packaged under sanitary conditions in accordance with good manufacturing practice. The product shall be processed in establishments meeting the sanitary requirements of the applicable Government inspection agency.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this specification. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation, and related services shall be borne by the contractor.

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4.3 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.4 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1 Ingredient examination. Conformance of ingredients to identity and condition requirements shall be determined by examination of pertinent labels, markings, U.S. Grade certificates, invoices, certificates of analysis, or other valid documents. In addition each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient lot or of any involved product.

4.5 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or other valid documents shall be cause for rejection of the involved product.

4.6 Tray pack inspection. The inspection lot shall include tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.6.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 108 ounces or if the net weight of any individual can is less than 106 ounces, the lot shall be rejected.

4.6.2 Product inspection. The finished product shall be inspected for conformance to the grade, vegetable proportions and drained weight requirements in accordance with the provisions of the United States Standards for Grades of Canned Peas and U.S. Standards for Grades of Canned Mushrooms and Regulations Governing Inspection and Certification of Processed Fruits and

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Vegetables, Processed Products Thereof, and Certain Other Processed Food Products. Product not equal to or better than the approved preproduction sample (see 6.1) in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

4.7 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

4.8 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: National Stock number, item description, contract number or date of pack markings missing, incorrect, or illegible. Minor: (1) Other required markings missing, incorrect or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) Reinforcement with non-metallic strapping or tape is not used; (4) Height of box liner less than inside depth dimension of box by more than 1/8 inch or extends above top edge of box.

4.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 108 ounces of food product, sealed, and thermoprocessed as specified in 3.4 and 3.5. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 90 pounds per base box. The entire inside area of the can shall be coated. The cans shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C. The product shall be packaged as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided

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with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pads shall be full length and width and be fabricated of grade V3c fiberboard. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be placed between cans and on the top and bottom of the stacked cans. The pads shall be full length and width and be fabricated of the same material as the box.

5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497. Paper labels are not permitted. Cans shall show the following statements:

TO HEAT

a. To heat in water: Submerge unopened can in boiling water. Simmer gently 15 minutes. Remove from water and open lid for serving.

b. To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place to prevent moisture loss. Place in a 350°F oven for approximately 15 minutes until contents reach an internal temperature of 165°F.

WARNING: Do not place closed can in oven as heating will increase internal pressure which may cause the can to burst.

YIELD

Approximately 25 servings of 1/2 cup (4 ounces) each.

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5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary markings in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Provisions for approved preproduction samples (see 3.6.1 and 6.3).
d. Level of packaging and packing required (see 5.1 and 5.2).
e. Type and class of unit load when unit loading is required (see 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection, selection, and approval of the first article.

Preparing activity:

Army - GL

Project No. 8915-A648

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
HHH-V-002133(Army-GL)	Peas And Mushrooms, Thermostabilized, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation:

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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